NOVA SCHOOL OF SCIENCE & TECHNOLOGY

DEPARTMENT OF COMPUTER SCIENCE

GONÇALO GOMES RODRIGUES

Master in Software Engineering Degree

USE OF DIFFERENTIAL PRIVACY FOR UNOBSERVABLE PRIVACY-PRESERVED COMMUNICATION

SOME THOUGHTS ON THE LIFE, THE UNIVERSE, AND EVERYTHING ELSE

MASTER IN COMPUTER SCIENCE AND ENGINEERING

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MASTER IN COMPUTER SCIENCE AND ENGINEERING

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Use of Differential Privacy for Unobservable Privacy-Preserved Communication
Some thoughts on the Life, the Universe, and Everything Else

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However, without any intention of conditioning the form or content of this text, I would like to add that it usually starts with academic thanks (instructors, etc.); then institutional thanks (Research Center, Department, Faculty, University, FCT / MEC scholarships, etc.) and, finally, the personal ones (friends, family, etc.).

But I insist that there are no fixed rules for this text, and it must, above all, express what the author feels.

"You cannot teach a man anything; you can only help him discover it in himself."

— **Galileo**, Somewhere in a book or speach
(Astronomer, physicist and engineer)

Abstract

The internet has become an indispensable tool for accessing information and exercising freedom of speech, a fundamental right that empowers individuals worldwide. However, authoritarian regimes and government agencies tend to suppress this right through surveillance and censorship by deploying increasingly sophisticated techniques, such as traffic analysis attacks, to monitor and undermine anonymity, placing users' privacy at significant risk. In response to the growing demand for secure and private communication, anonymity networks like Tor have emerged as essential tools for safeguarding online privacy and fighting censorship, which millions of people rely on daily. Yet, vulnerabilities in Tor's design, particularly against fingerprinting and correlation attacks, expose critical weaknesses that break its effectiveness.

This dissertation addresses these challenges by strengthening Tor Network's defences against the abovementioned attacks. The main goal is to develop an innovative solution that enhances Tor's existing architecture by introducing formal privacy guarantees. The solution is designed to maintain practical levels of throughput and latency, ensuring compatibility with real-world usage scenarios while providing users with adaptable and robust privacy protections.

At the core of the proposed solution is the integration of Differential Privacy principles to introduce carefully calibrated noise into traffic flows to protect users' privacy, which mitigates correlation attacks and empowers users with customizable privacy controls, allowing them to balance privacy and utility according to their needs. To the best of our knowledge, this work represents a pioneering effort in incorporating Differential Privacy into the Tor Network. It delivers a formal, scalable, and effective mechanism that bolsters online privacy, resists state-of-the-art censorship techniques, and upholds the fundamental right to free expression in an increasingly monitored digital world.

Keywords: Anonymity Networks, Differential Privacy, Privacy-Preserving Communication, Covert Communication, Traffic Analysis Attacks

Resumo

A Internet tornou-se uma ferramenta indispensável para aceder à informação e exercer a liberdade de expressão, um direito fundamental que dá poder aos indivíduos em todo o mundo. No entanto, os regimes autoritários e as agências governamentais tendem a suprimir este direito através da vigilância e da censura, recorrendo a técnicas cada vez mais sofisticadas, como os ataques de análise de tráfego, para monitorizar e comprometer o anonimato, colocando a privacidade dos utilizadores em risco. Em resposta à crescente procura de comunicações seguras e privadas, as redes de anonimato, como o Tor, surgiram como ferramentas essenciais para salvaguardar a privacidade em linha e combater a censura, nas quais milhões de pessoas confiam diariamente. No entanto, as vulnerabilidades na conceção do Tor, particularmente contra ataques de fingerprinting e de correlação, expõem fraquezas críticas que quebram a sua eficácia.

Esta dissertação aborda estes desafios, reforçando as defesas da rede Tor contra os ataques acima referidos. O principal objetivo é desenvolver uma solução inovadora que melhore a arquitetura existente do Tor, introduzindo garantias formais de privacidade. A solução foi concebida para manter níveis práticos de débito e latência, assegurando a compatibilidade com cenários de utilização reais, ao mesmo tempo que fornece aos utilizadores protecções de privacidade adaptáveis e robustas.

No centro da solução proposta está a integração dos princípios da Privacidade Diferencial para introduzir ruído cuidadosamente calibrado nos fluxos de tráfego para proteger a privacidade dos utilizadores, o que mitiga os ataques de correlação e dá aos utilizadores controlos de privacidade personalizáveis, permitindo-lhes equilibrar a privacidade e a utilidade de acordo com as suas necessidades. Tanto quanto é do nosso conhecimento, este trabalho representa um esforço pioneiro na incorporação da Privacidade Diferencial na rede Tor. Fornece um mecanismo formal, escalável e eficaz que reforça a privacidade online, resiste a técnicas de censura de última geração e defende o direito fundamental à liberdade de expressão num mundo digital cada vez mais monitorizado.

Palavras-chave: Redes de Anonimato, Privacidade Diferencial, Comunicação Preservadora de Privacidade, Comunicação Encoberta, Ataques de Análise de Tráfego

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Introduction

The internet has become an integral part of the daily life for most of the world's population. However, in some regions, individuals are denied the ability to freely share their stories or access vital information due to oppressive conditions and restrictions on freedom of speech. This oppression and restrictions are often enforced by totalitarian regimes that have vast resources and power to control the flow of information. [3, 60, 2]

As the demand for anonymity networks continues to expand, so does the necessity for robust privacy guarantees to safeguard secure and confidential communications. At the same time, authoritarian regimes intensify their efforts to restrict access exclusively to state-approved content, employing advanced measures to enforce their censorship policies and maintain control over information flow.

In order to enforce and control the network traffic, these regimes resort to the use of traffic analysis techniques to passively monitor the traffic and infer information about the users. Traffic analysis attacks are a significant threat to anonymity networks, as they can be used to infer users' activities and deny their privacy, as well as deanonymize them. [7, 56, 25] Some have demonstrated that advances in machine learning lead to more sophisticated and effective traffic correlation attacks, such as DeepCorr [34] and DeepCoFFEA [38], which leverage deep learning techniques to correlate traffic patterns and infer users' activities with high accuracy, in anonymity networks like Tor. [28]. Statistical Disclosure attacks also pose a threat to anonymity networks, as they identify sender-receiver pairs in anonymity networks. [12, 32] Website Fingerprinting attacks represent a major threat to anonymity networks, as they allow attackers to learn information about user's online activities, even on encrypted networks like Tor, by eavesdropping and collecting side-channel information from the network traffic, between the user and the entry node. With this information, the attacker can use machine learning techniques to infer the user's online activities, such as the websites visited, by analyzing the traffic patterns, with high accuracy. [45, 41, 9]

As these attacks become more sophisticated and effective, journalists, whistleblowers and any other person trying to express their freedom of speech and access information in these regions, are at risk of being identified and persecuted, leading to the decrease of this networks utility, mistrust and fear. Therefore, it is critical that anonymity networks

become more resilient to these attacks and provide stronger privacy guarantees to their users.

1.1 Motivation

Tor, one of the most popular anonymity networks, is an anonymous communication service based on the Onion Routing Protocol. [13] Tor is designed to protect users' privacy and security by routing their traffic through a network of volunteer-operated servers, encrypting it at each step, and ensuring that no single server knows both the source and destination of the traffic.

Anonymity networks remain vulnerable to attacks such as traffic analysis, which threatens the privacy and security of the users these networks are designed to protect. [7, 56, 25, 12, 32, 45, 41, 9] To address this challenge of circumventing censorship, researchers have proposed a range of solutions. Among these solutions, we discuss traffic encapsulation, fully encrypted protocols, and traffic shaping techniques.

Traffic encapsulation is a method that conceals data by embedding packets within other data formats or protocols, thereby obfuscating their true content. This technique leverages protocols, such as those for web browsing and video streaming, to disguise network traffic, making it indistinguishable. [52, 55, 5, 16, 21]

Traffic Shaping is another technique that aims to regulate data transmission to optimize network performance and to prevent timing and correlation attacks, contributing for privacy-prevention. Similarly, traffic shuffling is a technique that focus on reordering packets, with statistical or cryptographic techniques, to make it difficult for an attacker to correlate the traffic. [42, 30, 10, 40, 57]

Researchers have proposed various solutions. Hopper and Vasserman demonstrated that K-Anonymity is effective against traffic analysis attacks, such as Statistical Disclosure and long-term intersection attacks. [20] Nunes et al. proposed BriK, a Tor pluggable transport, that ensures uniform traffic patterns to make users indistinguishable. [37] Some work as Walkie-Talkie [53], Supersequence [54] and Glove [36] also use traffic shaping for defending against website fingerprinting attacks. Wright, Coull, and Monrose proposed Traffic Morphing [58], a traffic shaping technique that uses a Markov Chain to generate traffic patterns that are indistinguishable from the original traffic.

1.2 Problem

Although anonymity networks are designed to provide robust privacy and security, advancements in traffic analysis and correlation techniques have increasingly exposed vulnerabilities, reducing their reliability and trustworthiness for users—particularly in regions where pervasive surveillance by authoritarian regimes is a critical concern. These regimes actively invest in sophisticated surveillance methods and traffic manipulation strategies to maintain control over digital communication.

Tor, despite being one of the most widely used anonymity networks, is not immune to these challenges. It remains vulnerable to several attack vectors, including traffic correlation, congestion-based attacks, timing-based correlations, and fingerprinting techniques. [7, 28, 56, 25] Such vulnerabilities highlight the ongoing arms race between those striving to safeguard online anonymity and those seeking to undermine it through technical and infrastructural advancements.

The absence of standardized privacy metrics for evaluating privacy guarantees in a formal and rigorous manner raises concerns about the effectiveness of anonymizing networks like Tor. Research has demonstrated vulnerabilities in the Tor network to various attacks, including traffic analysis. For instance, Chakravarty et al. showed that statistical correlation techniques can be used to perform successful traffic analysis attacks on the Tor network. Additionally, Shi and Matsuura demonstrated that user privacy can be compromised through fingerprinting attacks.

1.3 Goals

The goal is to add differential privacy (DP) mechanisms into the Tor open-source project to preserve its existing security guarantees while enhancing its privacy protections. By incorporating DP, the aim is to provide quantifiable and adjustable privacy guarantees to users, addressing vulnerabilities to correlation attacks and improving resistance to traffic observability. Furthermore, the project seeks to analyze the trade-offs between privacy and utility within the Tor context and to evaluate the performance of the proposed solution compared to the original Tor network.

As stated in Section, it is important to ensure that anonymity networks are secure against threats like artificial intelligence and machine learning, and to address the lack of standardized privacy metrics to evaluate the performance of anonymity networks. We aim to create and implement a new algorithm that can be used in anonymity networks that leverages differential privacy to ensure privacy and security with higher performance than existing solutions. We also aim to evaluate the performance of the algorithm and compare it with other anonymity networks to prove that this solution is a viable alternative to usual anonymity networks solutions. To test the algorithm and its performance, we intend to realize correlation attacks on the existing networks and on our prototype and compare the results. To test performance, we will measure the latency and throughput of the algorithm and compare it with other anonymity networks.

1.3.1 Contributions

This thesis tries to address the problem of ensuring privacy and security in anonymity networks while maintaining high performance. In particular, the contributions of this thesis are:

1. Design a new Differential Privacy Based Algorithm for Privacy Preserved Communication in Anonymity Networks.
2. Implement the algorithm on top of the Tor network and provide a proof of concept.
3. Evaluate the performance of the algorithm and compare it with other anonymity networks; This contribution aims to prove if this solution is a viable alternative to usual anonymity networks solutions and if, as mention before, can allow users to switch from traditional to anonymity networks.

1.4 Report Organization

The remainder of this report is organized as follows: Chapter introduces anonymity networks and differential privacy, as well as some other related topics. The proposal of work is presented on Chapter, which provides some system model and guidelines for the development and discusses the proof method to verify the work. Finally, workplan is explained and addressed in Chapter

2

Related Work

2.1 Anonymity Networks & Mixnets

Mix networks (Mixnets) were introduced by Chaum [8] as a public key cryptography based protocol designed to obscure the relationship between senders and receivers in a communication network, as well as its content, without the need for a universal trusted authority. Mixnets consist of a series of intermediary servers, called 'mixes', through which messages are decrypted, encrypted, randomly permuted and sent forward. To transmit messages, participants encapsulate their data within successive layers of encryption, with each layer corresponding to a specific mix in the network — a method conceptually akin to onion routing [17]. The length of the encrypted messages is proportional to the number of mixes. Upon receiving a message, each mix removes the outermost layer of encryption, processes the decrypted instructions, and forwards the partially decrypted message to the subsequent mix. This iterative procedure continues until the message arrives at the final mix, which removes the last layer of encryption and transmits the plaintext message to the intended recipient. By design, this system ensures that no single mix possesses complete knowledge of both the sender's identity and the recipient's address, thereby achieving a robust level of anonymity.

Various designs of mixnets have been proposed, based on Chaum work, addressing multiple security and performance issues. Flash Mixing, introduced by Jakobsson [22], focuses on achieving strong anonymity guarantees with reduced latency and computational overhead by shuffling messages, broadcasting the encrypted list to all mixes, who then together compute the output. Hybrid Mixnets [23, 39] efficiently combine public-key and symmetric-key cryptography. Real-time Mixnets [27] aim voice and data communication where continuous data streams have to be transmitted and provide low-latency communication, as long as a certain delay at the start of a connection is tolerable. Work on this field has focused both on security and privacy guarantees, as well as on performance and scalability, with the goal of providing a practical and efficient solution for anonymous communication. Mixmaster [33] allows the sender of a message to remain anonymous to the recipient. Mixminion [11] uses fixed sized messages and supports anonymous replies

and ensures forward anonymity using link encryption between nodes. Onion Routing [17] followed a similar approach, but focused on low latency communication, where messages are encrypted in layers and decrypted by a chain of authorized nodes.

Tor [13] is one of the most popular low latency anonymity system and protects against sender-receiver message linking against a partially global adversary and ensures perfect forward secrecy, messages' integrity, and congestion control. However, Tor is vulnerable to traffic analysis attacks, in case the adversary is able to observe the entry and exit points of the network.

Mix-In-Place (MIP) [35] is a mixnet that uses a cascade of functions in a single proxy, instead of multiple intermediary nodes, to provide anonymity, and proving more resistant to traffic analysis attacks.

Vuvuzela [19] operates in rounds, leading to offline users' inability to receive messages and all messages must transverse a single chain of relay servers. This design protects against both active and passive adversaries unless there is no honest mix node in the network.

2.1.1 Groove

Groove [4] is a scalable, metadata-private messaging system designed to support users with multiple devices, enabling them to send messages at any time, even when recipients are offline, while conserving bandwidth and energy. Built on mixnets, Groove ensures unlinkability between senders and their messages by shuffling batches of messages across servers. Traditional mixnets, however, require all users to submit messages during every round and receive messages at a synchronized rate to prevent correlated traffic patterns, which limits their scalability and usability. To overcome these limitations, Groove introduces a novel approach called oblivious delegation. In this model, users interact with an untrusted service provider that participates in the mixnet on their behalf and synchronizes their clients across all devices. This ensures that even if the service provider is compromised, an adversary cannot infer communication metadata. Unlike prior systems such as Karaoke [30], Stadium [51], and Vuvuzela [19], which require users to be online simultaneously, communicate in synchronized rounds, and support only single-device usage, Groove addresses these restrictions, offering asynchronous messaging and multi-device support. A significant advantage of Groove lies in its ability to minimize the resource demands of each message channel on the mixnet, with a particularly notable reduction in memory usage compared to earlier systems. Groove ensures differential privacy even against attackers with full network control and the ability to compromise multiple servers. Users communicate through persistent message channels, similar to Tor's circuits, while their service provider handles message submission to the mixnet, stores received messages, and synchronizes clients across devices. To maintain anonymity, Groove leverages Parallel Mixnets, which efficiently scale with the number of servers by offering multiple parallel routes for processing messages. Differential privacy is a core goal

of Groove, ensuring that traffic patterns between a user's client and their service provider do not disclose information about the user's communication partners. This is achieved through a scheduler that operates independently of the sender-recipient relationship. In contrast to Loopix [40], Groove maintains user privacy even if their service provider is compromised, eliminating the need for users to operate their own servers.

2.1.2 Loopix

Loopix [40] is a low-latency anonymous communication system offering bi-directional 'third-party' sender and receiver anonymity as well as unobservability. It employs cover traffic and Poisson mixing to provide strong anonymity guarantees and resist traffic analysis by a Global Network Adversary (GNA). The system ensures robust sender-receiver unlinkability against a Global Passive Attacker (GPA) capable of observing all network traffic between users, providers, and mix servers, even in cases where some mix nodes are compromised. Loopix also guarantees sender online unobservability under a corrupt provider and receiver unobservability under the condition of an honest provider. Consequently, the GPA cannot infer the type of transmitted messages, while intermediate nodes remain unable to distinguish between real messages, dropped cover messages, or loops of traffic generated by clients and other nodes. To protect against active attacks, mixes and clients employ self-monitoring mechanisms through self-injected traffic loops. These loops not only act as cover traffic to strengthen anonymity but also enhance sender and receiver unobservability. Similar to Groove, Loopix utilizes service providers to mediate access to the network, manage accounting, and enable offline message reception. These semi-trusted providers enforce rate limits, store messages for offline users, and facilitate message retrieval at a later time. Despite this reliance, Loopix is resilient against adversaries capable of observing all communications and conducting active attacks. A distinguishing feature of Loopix is its continuous operation, unlike Groove's deterministic round-based approach. Messages in Loopix can be retrieved at any time, ensuring users do not lose messages while offline. Furthermore, Loopix employs Poisson mixing, a simplified version of the stop-and-go mixing strategy [29], which introduces independent delays for each message, making packet timings unlinkable. Unlike circuit-based onion routing, where a fixed path is established, Loopix determines the communication path for each individual message independently, even when sent between the same pair of users. This approach enhances privacy and unpredictability. The Poisson mix operates as follows: mix servers listen for incoming packets, check for duplicates, and decode received messages using their private keys. Duplicates are discarded, and the next mix packet is extracted. Decoded packets are not forwarded immediately; instead, each packet is delayed based on a pre-determined delay specified by the source.

2.1.3 Stadium

Stadium [51] is a point-to-point messaging system that ensures metadata and data privacy while efficiently scaling its workload across hundreds of low-cost providers operated by different organizations. Stadium achieves its provable privacy guarantees through the use of noisy cover traffic and differential privacy, which bounds metadata leakage over time. The system employs a mixnet architecture, where user messages and noise messages are distributed among providers and mixed in parallel. Each provider processes only a fraction of the total messages, ensuring scalability and efficiency. As long as at least one provider remains honest, Stadium achieves global verification and secure mixing. Communication in Stadium occurs in fixed rounds, with each round processing a batch of messages accumulated from users during the preceding interval. To enforce differential privacy, servers generate cover traffic at the start of each round. Noise messages are injected into the system in a single, large step before mixing begins. However, this design leaves room for malicious servers to discard noise messages before mixing. To address this vulnerability and ensure that noise remains in the system, Stadium employs cryptographic techniques for verifiable message processing. These techniques enable honest servers to verify the actions of others, preserving the system's differential privacy guarantees. To optimize the computational workload of this verification process, Stadium introduces hybrid verifiable shuffling. Stadium's mixing process relies on a parallel mixing scheme. Each server initially processes a small fraction of input messages, mixes them, and redistributes the mixed messages among other servers. This cycle of mixing and redistribution is repeated multiple times. Although messages are initially partitioned based on their originating server, the repeated cycles of splitting and mixing result in global mixing across all servers.

2.1.4 Problems & Analysis

Mixnets have been analyzed in terms of security and privacy guarantees, as well as performance and scalability. Zhu et al. revealed that flow-correlation attacks can be used to de-anonymize users in mixnets, in relation to the system parameterization, such as sample size, noise level, payload flow rate, and detection rate. [59]. Shmatikov and Wang demonstrated that mix networks are vulnerable to timing analysis, even with defenses in place, by the use of advanced statistical techniques, such as cross-correlation and machine learning, successfully de-anonymizing users. [44].

2.2 Tor Network

Tor is a circuit-based low-latency anonymous communication service [13] based on *onion routing* that aims to anonymize TCP-based applications, such as web browsing, SSH and instant messaging. The network relies on *Onion Routers* (ORs) that are responsible for maintaining TLS connections to every other OR and for forwarding traffic along the circuits, and are also maintained by volunteers.

An *Onion Proxy* (OP) is the local software run on the client and that handles connections with the users' applications, fetches the current network information and the lists of *Onion Routers* (ORs) and builds circuits through the network. Each OP maintains TLS connections to nodes they have been in contact recently and a pair of keys: long-term identity key and short-term onion key. Long-term keys are used to sign TLS certificates and its router descriptor. Short-term keys are used to decrypt requests, negotiate ephemeral keys and set up circuits.

Traffic travels the network through 512 bytes fixed-size *cells* with a header and a payload. The header contains the circuit identifier and the command to describe what to do with the payload. Based on the header's command, cells are either control cells or relay cells. Control cells are used to manage circuits and connections, while relay cells are used to carry data and can only be sent after a circuit is established. Relay cells have a digest assigned by the sender and its header and payload are iteratively encrypted with the symmetric key of each hop up to the target OR. This way, only the last OR in the circuit can read the payload, given the fact that the digest is encrypted to a different value at each step.

The sequences of ORs which traffic is routed through are called *circuits*, normally composed by 3 relays (entry or guard, middle and exit). To create a new circuit, a user's OP must execute a Diffie-Hellman key exchange with each OR in the circuit, one hop at a time. This way, each OR knows the previous and the next node in the circuit, but not the source and destination of the data. Once the circuit is established, relay cells are sent through the circuit,

2.2.1 Tor Circuit Scheduling

Traffic handling in Tor involves several buffers and schedulers. Each circuit is used by only one client but if multiple circuits use the same two ORs, they share the same connection. That is, OR will have multiple simultaneously connections with other ORs but only one to any given OR, and each connection will transport data for various circuits. When a OR receives a packet from an TCP stream, the packet gets demultiplexed, together with other possible incoming packets from different TCP streams, and they get placed into the kernel socket input buffers. Here packets are processed by the OS, usually in FIFO order, and sent to the Tor input buffers. Upon receiving the packets, Tor will remove the TLS layer and onion-encrypted (or onion-decripted, depending on the circuit's direction) and finally enqueued in the Tor Circuit Queue. Each relay maintains a queue for each circuit that it serves. Cells from the same Tor input buffer might not be enqueued in the same circuit queue, as they might belong to different circuits. The cells are then selected, dequeued, onion-encrypted and stored in a Tor output buffer. The data is then written to a kernel socket output buffer when the Tor output buffer contains sufficient data to form a TLS packet.

Tor has a congestion control mechanism (Circuit-Level Throttling), designed to regulate

communications across circuits, which ensures resource distribution, prevents network congestion and provides safety against attacks such as Denial-of-Service. As Tor is a low latency network and its security guarantees rely on how many users are using the network, it is important to have a good congestion control mechanism and practical levels of latency and throughput. However, it has been shown that Tor's Circuit Scheduling Algorithm allows busy circuits to crowd out bursty circuits, leading to congestion and latency issues [50]. Additionally, Jansen et al. have demonstrated that Tor's congestion occurs in the kernel socket buffers. Both these works have proposed solutions to improve Tor's congestion control and latency issues.

Even with the improvements in congestion control, as merging Tang and Goldberg and Jansen et al. works, Tor is still vulnerable to traffic analysis attacks such as website fingerprinting and correlation-based attacks [26]. Even though none of these works presented additional security risks or vulnerabilities to the Tor network, they have shown little or no improvement in the network against these attacks.

2.2.1.1 Exponential Weighted Moving Average (EWMA)

Tang and Goldberg proposed a circuit scheduling algorithm that handles circuits by their recent activity, where bursty circuits have higher priority over busy ones [50]. Generally, bursty circuits are those used for web browsing and instant messaging, and also referred as interactive streams. On the other hand, busy circuits are those used for file transfers, and also referred as non-interactive streams. Interactive streams are more sensitive to latency while non-interactive streams tolerate higher delays. The approach is based on the Exponential Weighted Moving Average (EWMA) algorithm, which assigns a weight to each circuit based on the number of cells sent on each circuit. When selecting the circuit to process, the algorithm chooses the one with lowest EWMA value. Usually, newly created circuits have a lower EWMA value, leading to a higher priority. Even though this algorithm has merged into Tor, it has been demonstrated that it actually reduces performance for clients under some network conditions [24]. Additionally, this work does not address the adversaries' ability to correlate and fingerprint users' traffic, which is a major issue in Tor network.

2.2.1.2 Kernel Informed Socket Transport (KIST)

Motivated by the still congestion problem with Tor and the lack of understanding of where this problem occurs, Jansen et al. found that the congestions occurs inside the kernel socket buffers and on Tor's sockets management. They proposed a new scheduling algorithm, Kernel Informed Socket Transport (KIST), that solves these problems by choosing from all circuits with writable data rather than just those belonging to a single TCP socket and by dynamically managing the amount of data written to each socket based on real-time kernel and TCP state information that can be queried from user space. This way, KIST reduces Tor's circuit congestion by more than 30%, reduces network latency by 18%, and

increases network throughput by nearly 10% [26]. KIST was merged and configured as default socket scheduling algorithm in Tor since January 2018, replacing the EWMA-based algorithm, referred in Section 2.2.1.1. Although, KIST authors acknowledge that it does not affect the adversaries abilities to collect accurate measurements required for the throughput correlation attack, compared to the 'vanilla' Tor scheduling algorithm.

2.2.2 Tor Vulnerabilities

Studies have pointed that Tor is susceptible to traffic analysis attacks, where an adversary can infer information about a user, like who is communicating with whom, by observing the traffic patterns. Here we cover some of the most common traffic analysis attacks against Tor, which have been demonstrated to be effective in de-anonymizing users.

2.2.2.1 Website Fingerprinting Attack

Chakravarty et al. also proposed a traffic analysis attack model against Tor where the adversary uses statistical correlation over the server to exit and entry to client traffic to find similar traffic patterns [7]. This attack consists on clients generating sustained traffic for a long period of time, by downloading a file for example, and the adversary being therefore able to inject traffic patterns by perturbing the TCP connection. This way, the adversary can obtain traffic analysis from the server to exit node and correlate it with the traffic from the entry node to the client, by finding the flow which carries the injected fingerprint. Testing showed that the attack was able to correctly identify the source of anonymous traffic 81.4% of the time. Deep Fingerprinting [45] is a website fingerprinting designed using deep learning techniques, that uses a simple input format and does not require handcrafting feature for classification. The authors demonstrated that this attack is 98.3% accurate in identifying the website visited by the user. Sirinam et al. proposed a website fingerprinting attack that uses triplet networks and a machine learning technique, that requires few training samples, called N-shot learning, which reduces the effort of gathering and training with large datasets [46]. Triplet networks. Finally, this work achieved up to 95% accuracy in identifying the website visited by the user, only by using 20 examples per website.

2.2.2.2 Correlation-Based Attacks

Correlation attacks were acknowledged by Dingledine, Mathewson, and Syverson [13] as a potential threat to Tor's anonymity. These attacks exploit the adversary's ability to observe both the entry and exit points of the network, allowing them to correlate the traffic patterns and de-anonymize users. Sun et al. [47] proposed asymmetric traffic analysis as an end-to-end timing analysis that allows AS-level adversaries to compromise Tor users' anonymity. Internet path from exit relays to the web server may differ from the path from the web server to the exit relay, allowing the adversary to observe the TCP

acknowledgement traffic on the path from the server to the exit relay, even if the adversary is enabled to observe the data traffic on path from the exit relay to the server. This attacks might be applicable where the adversary observes: data traffic from the client to the entry relay and from the exit relay and the server; data traffic from the client to the entry relay and TCP acknowledgement traffic from the server to the exit relay; TCP acknowledgement traffic from the entry relay to the client and data traffic from exit relay to the server; or TCP acknowledgement traffic from the entry relay to the client and TCP acknowledgement traffic from the server to the exit relay. The authors proposed a suite of new attacks against Tor called 'RAPTOR', where they demonstrated an accuracy of 95% in correlating client/server pair.

DeepCorr [34] and its extension DeepCoFFEA [38] are both systems that leverage advanced deep learning techniques to correlate traffic. DeepCoFFEA [38]

2.3 K-anonymity

To address the exponential growth in the number and variety of data collections containing person-specific information, and the pressing need to prevent privacy compromise, *k-anonymity* [48] emerged as a foundational privacy model. This model ensures data anonymity not only by removing explicit identifiers such as names, addresses, or phone numbers but also by protecting against re-identification through linking or matching data with external sources. It achieves this by mitigating the risk posed by unique characteristics within the dataset.

Earlier work by Sweeney demonstrated the vulnerability of anonymized data to linkage attacks using the 1990 U.S. Census. They highlighted that 87% of the U.S. population reported characteristics that rendered them unique, and over half of the population could be uniquely identified using just three attributes: place, gender, and date of birth [49]. To counter these risks, Sweeney defined *k-anonymity* as a property satisfied by a dataset `D` if, and only if, each sequence of values for quasi-identifiers in `D` is indistinguishable from at least $k-1$ other records in `D`. In practical terms, any tuple in `D` must be identical to at least $k-1$ other tuples concerning its quasi-identifiers. This property ensures that the data cannot be easily linked to other sources and protects users from re-identification, with the privacy guarantees strengthening as k increases.

Ahn, Bortz, and Hopper extended the concept of *k-anonymity* to communication systems, defining a protocol as sender k-anonymous if it ensures that an adversary attempting to identify the sender of a message can narrow their search to a set of k potential senders. Similarly, receiver k-anonymity guarantees that an adversary can only narrow down the possible recipients to a group of k [1].

Despite its foundational role, *k-anonymity* remains vulnerable to several types of attacks, including unsorted matching, complementary release, and temporal attacks, as noted by Sweeney [48]. Unsorted matching attacks exploit the order of tuples in a released dataset, which can be mitigated by randomizing the tuple order. Complementary release

attacks arise when subsequent data releases are combined, allowing adversaries to re-identify individuals by correlating datasets. Temporal attacks exploit data changes over time, such as additions, deletions, or modifications, leading to violations of k-anonymity guarantees as datasets evolve.

Further research has explored enhancing k-anonymity to address its limitations. For instance, Hopper and Vasserman observed that receiver k-anonymity improves effectiveness against long-term intersection and statistical disclosure attacks. Additionally, incorporating periodic sender k-anonymity under low churn conditions enhances resistance to mass surveillance, albeit without achieving an optimal cost function [20].

Recognizing that k-anonymity alone may not suffice to protect privacy in the era of big data, Gosain and Chugh proposed combining k-anonymity with Differential Privacy. This hybrid approach addresses the challenges posed by the massive scale and interconnected nature of modern data systems [18]. Similarly, in response to the proliferation of social networks and the accompanying surge in data collection, Campan and Truta introduced an anonymization technique for social network data, masking it according to the k-anonymity model [6].

While k-anonymity offers practical privacy guarantees, it is primarily suited for systems prioritizing efficiency over robust privacy protections. It remains less applicable in scenarios where strong, comprehensive privacy guarantees are essential.

2.4 TorKameleon & Stream Covered Channels

2.5 Differential Privacy

Differential Privacy (DP), introduced by Dwork, addresses the critical challenge of enabling the extraction of meaningful insights about an underlying population while rigorously safeguarding individual privacy. This framework ensures that user data can be shared and analyzed without exposing individuals to adverse consequences, regardless of the presence of auxiliary information or other datasets [14]. The intuition behind DP is that each row contains the data of a single individual, and the privacy goal is to simultaneously protect every individual row while permitting statistical analysis of the dataset as a whole.

Differential Privacy achieves its guarantees by introducing carefully calibrated randomness into the analysis process. Privacy arises from the principle of plausible deniability, ensuring that any specific outcome cannot be definitively attributed to an individual. At the same time, accuracy is preserved through a precise understanding of the noise generation mechanism. Randomization has a strong role [15]

Under this paradigm, any sequence of outputs remains "essentially" equally likely, irrespective of whether a particular individual's data is included or excluded. The probabilities of these outputs are governed by the randomized choices of the privacy mechanism, with the degree of privacy loss quantified by a parameter, ε. A smaller value of ε corresponds to stronger privacy guarantees, effectively balancing the trade-off between utility

and privacy.For a given computational task T and a given value of ε there will be many differentially private algorithms for achieving T in an ε-differentially private manner. When ε is small, finding a highly accurate ε-differentially private algorithm for T can be difficult, much as finding a numerically stable algorithm for a specific computational task can require effort.

There are two primary formulations of differential privacy:

- **(ε, 0)-Differential Privacy**: Ensures that, for every execution of a mechanism $M(x)$, the output is (almost) equally likely to occur across all neighboring datasets.

- **(ε, δ)-Differential Privacy**: Provides a relaxed guarantee, where for any two adjacent datasets x and y, the privacy loss is bounded by ε with probability at least $1 - \delta$. This formulation is useful for achieving approximate privacy guarantees in scenarios where perfect differential privacy is infeasible.

Differential Privacy offers several notable properties that make it uniquely robust: [15]

- **Protection Against Arbitrary Risks**: It extends privacy protection beyond traditional re-identification attacks, addressing broader privacy threats.

- **Neutralization of Linkage Attacks**: DP effectively mitigates risks posed by linkage attacks, regardless of whether they are based on past, present, or future datasets or auxiliary information.

- **Quantification of Privacy Loss**: Unlike binary privacy concepts, DP quantifies privacy loss, providing a measurable and interpretable framework.

- **Composability**: The cumulative privacy loss from multiple computations can be analyzed and controlled, ensuring consistent privacy guarantees across workflows.

- **Group Privacy**: DP allows for quantifying and managing the privacy loss of groups, extending its utility to collective data analysis.

- **Closure Under Post-Processing**: Differential Privacy is inherently robust to post-processing. That is, any function of the output of a differentially private algorithm, computed without access to the underlying private dataset, cannot weaken the privacy guarantee.

DP promises that the behavior of an algorithm will be roughly unchanged even if a single entry in the dataset is modified.It is a strength of differential privacy that composition is 'automatic', in that the bounds obtained hold without any special effort by the database curator.

Differential Privacy can be implemented in two primary settings: [14]

- **Non-Interactive Setting**: A trusted entity publishes a "sanitized" version of the dataset. This process often involves techniques such as data perturbation, sub-sampling, or removing identifiable attributes (e.g., names or social security numbers). These methods are aimed at minimizing the risk of re-identification.

- **Interactive Setting**: In this model, the trusted data collector provides an interface through which analysts can submit queries and receive responses. The answers are typically noisy, with the noise calibrated to ensure differential privacy while preserving utility.

The independent use of an ε-differentially private algorithm and an ε'-differentially private algorithm, when taken together, is $(\varepsilon + \varepsilon')$-differentially private.

2.5.1 Applications of Differential Privacy

Differential Privacy emerged as a solution for privacy-preserving data analysis

DP neutralizes linkage attacks: since being differentially private is a property of the data access mechanism, and is unrelated to the presence or absence of auxiliary information available to the adversary, access to an certain dataset does not increase the adversary's ability to perform a linkage attack on a certain user whose data is in another dataset, compared to someone whose data is not in the dataset.

2.6 Summary

3

Elaboration

3.1 System Model

3.2 Prototype Guidelines

3.3 Validation

3.4 Summary

4

Workplan

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